ZOO404

MID TERM SUBJECTIVE BEST OVER ALL FILES

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Define physiology?

Ans: It is the study of functions of tissues, organs and organ systems of animals to understand the mechanisms operating in animals.

Features of Cardiac Muscles. 3

1. Cardiac muscle cells (myocyte) are mononucleate.
2. Skeletal muscle cells are multinucleate.
3. They don't produce discrete PSP.
4. They only increase or decrease the strength of myogenic contractions.

Donnan Equilibrium

"If diffusible solutes are separated by a membrane that is freely permeable to water and electrolytes but totally impermeable to one species of ion, the diffusible solutes become unequally distributed between the two compartments".

Describe the types of feedback system?

1= Negative Feedback:(Negative feedback is a regulatory mechanism in which a 'stimulus' causes an opposite 'output' in order to maintain an ideal level of whatever is being regulated.)

2= Positive Feedback: Positive feedback is the amplification of a body's response to a stimulus.

What is meant by muscle twitch?

Ans: A muscle twitch is a single contraction in response to a single action potential.

Three Components of a Single Muscle Twitch: 1= Latent period, or lag phase 2= Contraction phase 3=Relaxation phase

What is sarcoplasmic Reticulum?

Ans: The sarcoplasmic reticulum (SR) is a membrane-bound structure found within muscle cells that is similar to the smooth endoplasmic reticulum in other cells. The main function of the SR is to store calcium ions (Ca2+).

Chemical Types of Hormones 2

On the Based on their structure and pathway for synthesis, hormones are divided into four groups:
1=Peptide and protein hormones
2=Amine hormones
3=Steroid hormones
4=Prostaglandins

Negative Feedback Systems;

1= The most common form of regulation
2= The end product works to stop or slow down the process

Spinal Motor Neurons diagram:

A motor neuron and the muscle fibers that it innervates form a motor unit.

Function of Parathormone (PTH):

Increases blood Ca2+ levels; decreases blood phosphate level

What is Acclimation?

Acclimation refers to a physiological change in an individual stimulated by exposure to a different, often stressful, environment. As such it represents physiological phenotypic plasticity.

How permeability can be altered?

Hormones and some other molecules can alter the permeability to certain substances. Hormones react with receptor sites and influence channel size or airier mechanisms.

Acclimatization

•An adaptive change within an individual animal.

Example: 1=An animal migrates to a high altitude area 2= Faces low oxygen partial pressure

Effects: 1=Lung ventilation rate increases initially 2=Ventilation drops to normal in few days 3=Animal has acclimatized to the high-altitude

Acclimation: Changes are induced experimentally in the laboratory

Example: 1=Simulating high-altitude conditions in lab 2=An animal physiologist places the animal in a hypobaric chamber.

Effects: 1= Breathing rate reacts in the same way. 2= Animal becomes acclimated to the experimental conditions

Rigor Mortis After death, human and other animal's bodies gradually become rigid.This condition is called rigor mortis.

Role of Ca2+ in Cross-Bridge Attachment

1= Ca2+ ions play a crucial role in regulating the contractile activity of muscles.
2= Calcium helps to expose the active sites of actin that bind myosin heads to form cross-bridges, necessary for contraction.

Neurofibrils: Delicate threads running from cell body into the axon and dendrites.

Neurofibrils Functions: 1=Give support and shape to the neuron. 2=Transport substances from cell body to the distal cell processes.

Quantal Release of Neurotransmitter 2

•Neurotransmitters are generally released in tiny packets called quanta.
•Each quantum may consist of about 2000 to 10,000 molecules of transmitter molecules.

Conformers and its type 2

Animals that conform to external changes. They Are unable to maintain homeostasis for internal conditions

Types: 1=Osmoconformers 2=Oxyconformers 3=Thermoconformers

Terminal cisternae of Sarcoplasmic Reticulum 3

Terminal cisternae are larger chambers that are closely associated with T-tubules (12 nm apart). These are the primary site of calcium release as they have channel proteins that open with AP in T tubules.

Action potentials: are the type of signals that neurons use to send information over long distances.

Properties of Hormones:

1= They act only on specific target cells or tissues. 2=Hormonal action depend on the type of receptor. 3= Hormones are active at very low concentration. 4= The amount of hormone produced by an endocrine gland is small. 5= It is further diluted in the blood and interstitial fluid.

Phases of AP: 1. Stimulation 2. Rising phase (Depolarization) 3. Peak phase. 4.Falling phase (Repolarization) 5. Undershoot (after hyperpolarization)

Synthesis of Steroid Hormones: 1=Steroid hormones are not stored in the glandular cells normally. 2=They are synthesized from cholesterol instantly when a stimulus is received. 3=For this purpose, cell maintains large stores of cholesterol which can be rapidly mobilized.

T Tubules: 1= T tubules are thin internal extensions of the cell membrane. 2= They are less than 0.1 μm in diameter. 3= They innervate the cell at the level of Z disk. 4= Make branching networks around the perimeter of each myofibril.

Role of T Tubules: 1= T tubule system provides the anatomic link between the surface membrane and the myofibrils deep inside the muscle fiber. 2=When an action potential spreads over a muscle fiber membrane, a potential change also spreads along the T tubules to the deep interior of the muscle fiber. 3= In the cell interior, T tubes are linked to SR. 4= SR releases Ca2+ that permit myosin cross-bridges to attach and generate force for contraction.

Resting Potential Values and Ions Responsible for RMP:

Resting Potential Values: 1= The non-excited or "resting" cell has a potential difference, Vrest across its membrane. 2= Values of Vrest vary in different types of cells. 3= Most neurons have an RMP of -70mV, when no impulse is being conducted.

Ions Responsible for RMP: 1= Plasma membranes are permeable to more than one type of ionic species. 2= All diffusible ions affect the potential across a membrane in proportion to the permeability. 3= Non-diffusible ion species have no effect on MP. 4= Major ions responsible for Vrest : K+, Na+ & Cl-.

What is meant by sliding filament theory?

H. E. Huxley and A. F. Huxley proposed this theory in 1954 to explain muscle contraction.

It states that the mechanism that allows muscles to contract. According to this theory, myosin (a motor protein) binds to actin. The myosin then alters its configuration, resulting in a "stroke" that pulls on the actin filament and causes it to slide across the myosin filament

What is Nissl's bodies?

Consist of a group of ribosomes and rough endoplasmic reticulum These bodies are associated with protein synthesis.

Name the hormones of Hypothalamus?

There are two categories of hypothalamic hormones: 1=Hypophysiotropic hormones 2=Neurohypophyseal hormones

What is dwarfism?

Insufficient secretion of growth hormone GH during childhood and adolescence causes underdevelopment of the body (short stature) which is known as dwarfism.

What is meant by calcitonin?

Calcitonin is a protein hormone secreted by the parafollicular cells of thyroid gland. It is released in response to hypercalcemia. It is an important hormone for calcium metabolism and calcium homeostasis.

Name the Laws which are obeyed by physiology?

Ohm's Law, Boyle's Law, Ideal Gas Law, Law of Gravity, Kinetic & potential Energy, law of Inertia, law of Momentum, Velocity, Drag

What is membrane Flux?

Membrane flux is defined as the daily or hourly water flow through a membrane's surface area (i.e. GPD/ft2), and the treated water that passes through the permeate. The flux rate depends on the membrane type and several physical and environmental operating conditions.

What is tetanus and name its types?

A sustained muscle contraction evoked by stimulation from simultaneous multiple impulses is known as tetanic contraction. Types: A tetanic contraction can be either: 1= Unfused (incomplete) 2= Fused (complete)

Write the three functions of hormones?

1= Maintain homeostasis 2= Coordinate body's responses to stress 3= Mediate responses to many environmental stimuli.

Name the three lobes of pituitary hormone?

1= Anterior lobe (adenohypophysis) 2= Intermediate lobe (pars intermedia) 3= Posterior lobe (neurohypophysis)

Write the names of tropic and non-tropic hormones of Adenohypophysis?

Tropic: Adenohypophysis secretes four tropic hormones: 1= Adrenocorticotropic hormone (ACTH) 2= Thyroid stimulating hormone (TSH) 3= Luteinizing Hormone (LH) 4= Follicle stimulating hormone (FSH)

Non tropic: Adenohypophysis secretes three hormones that act directly on their Target tissues. These are: 1= Melanocyte stimulating hormone (MSH) 2= Prolactin (PRL) 3= Growth hormone (somatotropin)

Define Chromaffin Cells?

The cells of adrenal medulla are modified postganglionic sympathetic neurons. They are known as chromaffin cells because they stain easily with chromium salts. Chromaffin cells that produce norepinephrine have dark staining irregular granules.

What is pancreas?

Pancreas is a dual gland having exocrine as well as endocrine roles. It is composed of two types of tissues: 1= Acini which are exocrine. 2=Islets of Langerhans which are endocrine.

Write down the phases of hormone action?

Acclimatization of hormone is divided into two phases: 1= Forming the hormone-receptor complex. 2= Producing the effect or response.

Write the functions of cell body?

Cell body is the Main nutritional part of the nerve cell. It is Responsible for synthesis of materials necessary for growth and metabolic maintenance of the cell.

Define Synapse and write name of its types?

Synapse are the Junctions between axon terminals of one neuron and the dendrites of another neuron. Types: Electrical synapse and chemical synapse

Where the electrical synapse occurs?

Electrical synapses are relatively rare and most signaling between neurons takes place through chemical synapses.

Electrical transmission has been discovered in Many cells of central nervous system, Retina of the eye, Smooth muscle fibers Cardiac muscle fibers, Most receptor cells and Some axons of PNS.

Difference between Thin and Thick filaments?

Thin filament: Thin filaments are a polymer of actin with tightly bound regulatory proteins troponin and tropomyosin.

The functions of the actin based thin filaments are interaction with myosin to produce force, the regulation of force generation in response to Ca2+ concentration and the transmission of the force to the ends of the cell.

Thick filament: Thick filaments composed of myosin they occupy the center of sarcomeres where they partially overlap with thin filaments.